Sid Diamond

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Profile

Recently completed an MSci in Physics at Imperial College London, focusing on uncertainty quantification in both simulations of complex systems and in transformer architectures. I also have software engineering experience leading development of an MVP multi-AI-agent planning application. Additionally, my recent work includes research on dataset bias in supervised learning, an independent project with Imperial's High Performance Computing group applying transformer-based NLP to geotagged text documents, and freelance technical AI safety research exploring the relationship between AI systems and their internal models of the world. Currently an Impact First Fellow at Arcadia Impact. Recently completed the BlueDot Impact intensive technical AI safety course. Currently BlueDot Impact rapid grant funded to research emergent dynamics in multi-agent interactions causally related to model identity.

Education

Integrated MSci in Physics 2022–2026
Imperial College London, UK
Overall: 2:1 (65%) — Master's Year: 2:1 (69%)

Relevant Modules:
- Object-Oriented Programming (98.70%)
- Information Theory (91.00%)
- Statistics (72.90%)
- Problem Solving (79.30%)
- Master's thesis on using simulation to predict the growth of complex systems (74.15%)
- Undergraduate thesis on uncertainty quantification in transformer-based NLP systems (69.50%)
- Computational Sociology: research on dataset bias in supervised machine learning algorithms (67.00%)
- Communicating Physics: educational policy report (84.10%, highest mark in cohort)

A-Levels 2014–2021
Stoke Newington School and Sixth Form, London, UK
Further Maths (A*), Maths (A*), Physics (A*), Chemistry (A*), EPQ (A*)
Awards: Chemistry Olympiad Silver (2021); Kelvin Prize Winner (2020, *Peterhouse College Cambridge University Science Competition*)

Research Experience

Rapid Grant Funded Research and Intensive Technical AI Safety Course Current
BlueDot Impact —Supervisor: Jason Tang
- Completed BlueDot Impact's Intensive Technical AI Safety course, furthering technical understanding of key AI safety research areas, including alignment and RLHF, mechanistic interpretability and evaluations
- Awarded a BlueDot Rapid Grant to investigate how model self-conception influences emergent behaviour and safety outcomes in multi-agent AI systems
- Designing an extension to Douglas et al.'s modification of the agentic-misalignment experiment, varying agent self-conception and population size to study emergent cooperation, persuasion, sabotage and harmful behaviour
- The active research: Proof of concept experiment

Master's Thesis: Network Growth in Finite-Sized Networks 2025–2026
Theoretical Physics Group, Imperial College London — Supervisor: Associate Professor Tim Evans
- Built and simulated a predictive model for connectivity growth in finite complex systems (such as networks of individuals and AI agents), with emphasis on empirical validation and model behaviour under realistic constraints (the effects of homophily and cumulative attachment)
- Created a statistical goodness-of-fit test to quantify confidence in the predictive model and assess agreement with observed network behaviour
- Produced research code and write-up that were validated by my supervisor and an external marking panel as a meaningful extension of existing literature; work is currently being used by visiting researchers
- Code and report: Master's thesis code and write-up

Hate Speech Detection Dataset Research 2025–2026
Advanced Humanities Department, Imperial College London — Supervisor: Dr Catherine Webb

- Conducted empirical research on how benchmark datasets can introduce bias into supervised machine learning systems for hate speech detection, with a focus on evaluation quality and downstream fairness implications
- Proposed a policy framework to mitigate harms arising from biased datasets and evaluation pipelines
- Report: On Hate Speech Detection

Uncertainty in Transformer-Based NLP Algorithms 2024

Independent Research & Undergraduate Thesis — Supervisor: Dr Aidan Crilly

- Worked with the Imperial High Performance Computing research group to develop a transformer-based NLP tool for topic extraction and document-level scoring of actionable policy insight in economic climate policy documents
- Implemented BERT-based modelling workflows and analysed model outputs for applied NLP tasks on real-world policy text. Achieved industry standard f1 scores across a range of datasets
- Extended the project into an undergraduate thesis on uncertainty quantification in transformers, investigating per-parameter variance estimates due to fine-tuning, overall model uncertainty using Monte-Carlo dropout, and information loss associated with dimensionality reduction. Only self-proposed project in cohort
- Code and write-up: Transformer Uncertainties in NLP

Professional Experience

Impact First Fellow 2026

Arcadia Impact

- Fellowship focused on high impact work such as AI safety, with emphasis on population ethics
- Independently researched simulating job displacement resulting from the release of frontier AI models
- EA Forum post: EA Forum post on using complex systems to model job displacement

AI Safety Researcher 2026

Dovetail Research — Supervisors: MATS Scholar Alex Altair and Dr Alfred Harwood

- Conducted paid freelance research for Dovetail Research studying the mathematical foundations of AI safety
- Investigated extensions of the Touchette–Lloyd (2003) theorem to better characterise mutual information between an agentic AI system's world model and the external world
- Contributed to theoretical work aimed at supporting more empirical measures of AI goal alignment
- Report: Agentic Foundations AI Research

Product Engineer Intern 2025

Mindlace Technologies

- Led development of an MVP agentic planning application, implementing full-stack product functionality in a fast-moving startup environment
- Built practical workflows for an AI-driven planning product, combining rapid prototyping with modern software tools and iterative product development
- Project link: Agentic AI Planning Application MVP

Science & Maths Tutor and Teacher 2021–2024

Self-Employed

- Taught A-level Physics and mathematics across a range of student ability levels, including a four-month placement independently teaching a 25 student GCSE physics class and a 10 student A-level physics class
- Privately tutored, alongside my university studies, students across a diverse range of abilities

Technical Skills

Programming: Python, TypeScript, NetLogo, Claude Code/ Codex

ML / NLP: PyTorch, HuggingFace Transformers, BERT, NLP pipelines, supervised learning, unsupervised learning, NumPy, SciPy, Pandas, Matplotlib, Plotly, dimensionality reduction, network analysis, graph theory, linear algebra, fine-tuning

Software & Prototyping: GitHub, LaTeX, Supabase, n8n, Shadcn/ui, Pinecone, OpenRouter, Firecrawl, Tavily, OpenAPI, AWS cloud, Microsoft Azure

Additional Activities

Sport: Imperial Men's Football 1st and 2nd teams; previously captain of the 2nd team. Also enjoy recreational running.

Journalism: Comment writer for *Felix* (Imperial College student newspaper), including articles on ESG and artificial intelligence.

Music: Guitar.